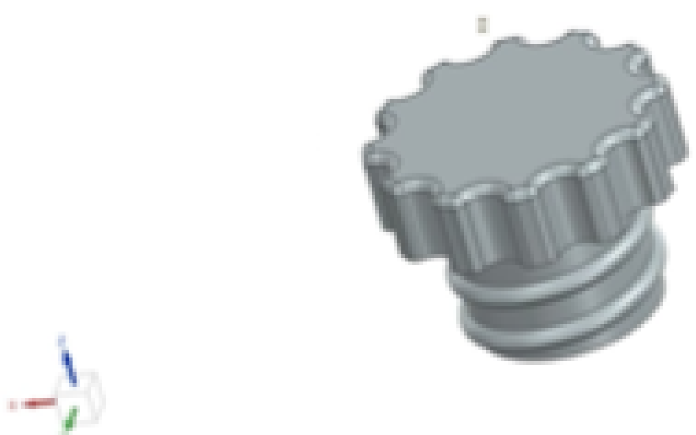
Vehicle Gas Cap

Created by Aiden Aerni



Summary

Product Description

- Aftermarket vehicle gas cap designed for a 2000 Jeep Cherokee
- Material: **6061 Aluminum**
- Chosen for durability and high-quality, glossy finish

Manufacturing Method

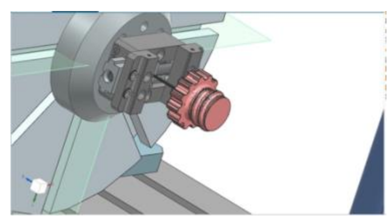
- Machine: **Haas VF4 4-axis CNC mill**
- Capable of accurate thread and sealing surface machining
- Fixture: **Compact milling vise with high riser**
- Ensures proper part orientation and rigidity during machining

Operations & Tooling

- Operation 1: Roughing of rotary features
    - **Tool: 8 mm, 2-flute end mill**
- Operation 2: Finishing operation
    - **Tool: 7 mm, 2-flute ball end mill**
- Designed to achieve final geometry and surface finish

Projected Output

- Total annual output: 520 units per year



Production Cost per Part

- Raw material cost: $31.68
- Machine cost (amortized): $3.06 per part
- Fixture cost (amortized): $0.50 per part
- Labor cost: Not included (owner-operated)
- Total cost per part: **$35.24**

Sales Price & Revenue

- Sale price: $70.00 per part
- Competitive with similar aftermarket products
- Annual sales revenue: $36,400
- Estimated annual profit: $18,075.20

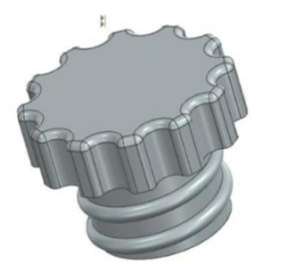
Sustainability Impacts

- Overall environmental **impact is relatively small**

- Largest impact occurs during the **use phase**
- **Recycling** at end of life helps reduce total footprint

Introduction
For my product I have chosen to design and manufacture a custom vehicle gas cap, like the one on the list of acceptable products. I have decided to keep the gas cap simple but with some personal touches of my own. In this report I will explain how a product like this may build a business. I will explain the essential topics like value proposition, manufacturing method, operations and tooling, projected output, production cost, sales and revenue, and sustainability impacts.

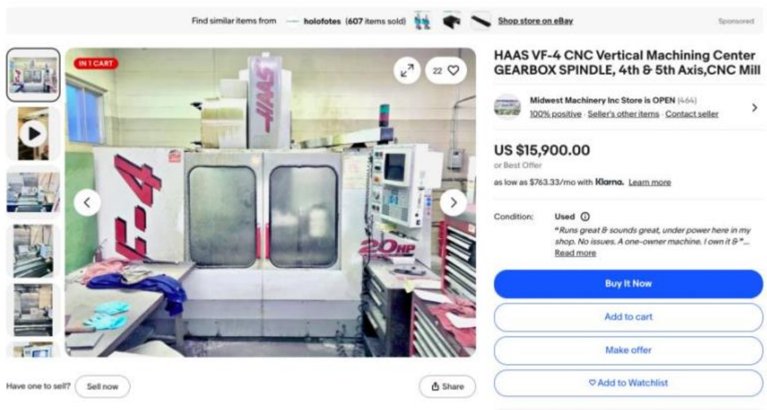
Product and Value Proposition
This product will serve as a solution for those who prefer personalized accessories to their vehicle rather than stock equipment. It will also be a solution for anyone who may have a defective gas cap and needs a replacement. A product like this will have to be sized per vehicle. This means that a standard gas cap size may have to be established amongst the business to keep producing efficiently. A custom engraved gas cap ranges from a price of $50 to $150 depending on the intricacy of the design and the material used for the product (taken from Amazon and various private machine shops). I expect to charge $70 per gas cap assuming the product has been manufactured with heavy material like steel or aluminum as compared to plastic. I plan to use T6061 Aluminum for my product. There is credible information from marketreportsanalytics.com which suggests there is a marketable demand for gas caps, and that selling this number of units is very achievable with a fuel filler/ fuel cap market of around 2 billion dollars per year. My product is shown in the first figure while the inspiration for the product in the next.



Manufacturing Method
Features of the Part
- The features of this part are as follows:
- Two cylinders, one as top end of the cap and the other which inserts into the gas tank A helical thread along the bottom cylinder
- Patterned circular cutouts along the top edge of the top end cylinder
- Chamfered edges along all patterned circular cutouts
- Chamfered ends of the thread

- A hole from the bottom end of cylinder extends up to the top end cylinder.

Machine Choice



A 4-axis CNC machine will be used to accurately machine the helical thread along the cylindrical portion of the gas cap. By orienting the part on its side, most of the features can be machined in a single setup. The only feature that cannot be completed in this orientation is the bottom hole of the cap, which will require a separate drilling operation with a different workpiece and fixture orientation. The machine under consideration offers 4- or 5-axis capability and includes a quick tool-change system. Compared to other machines with similar multi-axis functionality, this machine provides good value and is well suited for machining this part due to the range of orientations it can achieve. After accounting for installation and maintenance, the estimated total cost of the machine is approximately $32,000. The figure shows the machine of choice, the Haas-VF4 CNC Vertical Machining Center.

Orientations during machining

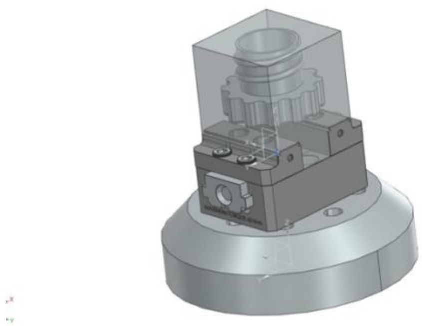
With the central axis of the part running from top to bottom end, the part will need to be oriented aligned with the z axis to machine the large hole, top end cylinder and its chamfered edges. It must also be rotated along the x axis to machine the threads and its chamfers.

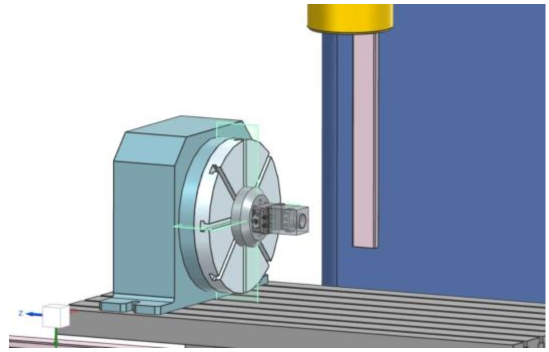
Degrees of Freedom

With the part constrained in the vise, all degrees of freedom are restricted. This includes rotational, and translational along the xyz planes. The part will rotate with the table and vise which is intentional

Fixturing Strategy

The part will be fixtured by a vise clamp, which will be attached to a riser component fixed to the rotating table. By keeping an offset of 30mm on my stock, I can ensure that all features of my part may be machined. The stock will be oriented so that the top end cylinder will be normal to the fixture. After machining, the part will have to be cut with a bandsaw to separate from the remaining stock, which is clamped in the feature, then a surface finishing process will be applied to the top face of the cap. Below are figures which show the fixturing strategies, and how the fixture will be oriented in the machine





Operations & Tooling

After finalizing the model and fixture strategy, I switched to the manufacturing application to create a 4-axis rotary milling program. I defined the machine coordinate system, stock, and fixturing, then programmed roughing and finishing toolpaths—including rotary contouring and 4-axis swarf operations—for machining the complex curved grip surfaces.

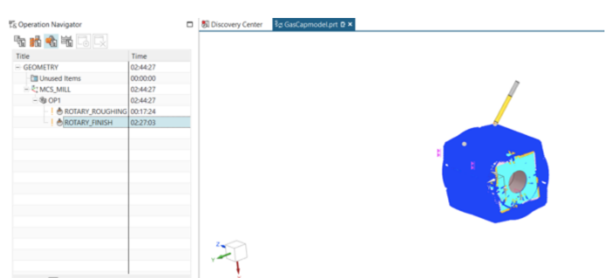
Speeds, Feeds, & Tooling

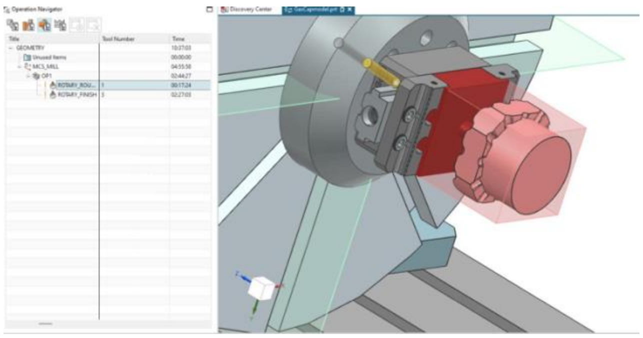
The machining process was set up with feed rates of 1200 mm/min and spindle speeds of 10,000 rpm, and the total machining time for the gas cap ended up being 2 hours and 45 minutes. After verifying all toolpaths with simulation, I generated the final NC program for manufacturing. Below are the tooling times for each operation, the first a roughing, and second a finishing operation.

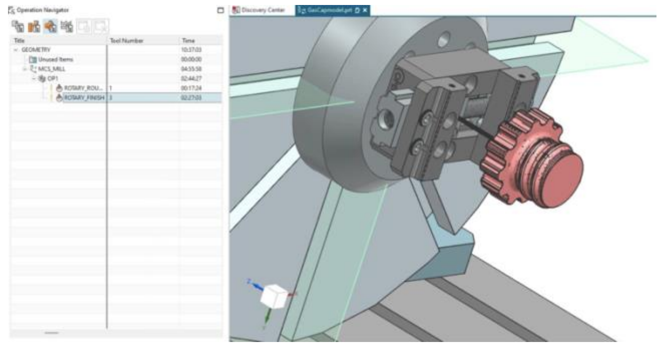
Tool Description	Description	Notes	Holder	Tool Number	Time
	mill_contour				02:44:27
	mill_contour				00:00:00
	MCS_MILL				02:44:27
	WORKPIECE				02:44:27
Milling Tool Custom Rou...	ROTARY_ROUGHING			1	00:17:24
Milling Tool-Ball Mill finish	ROTARY_ROUGHING			3	02:27:03

Tool Paths & IPWs

For the two operations, finishing and roughing, the tool paths were significantly different. This is to be expected as the finishing operation is significantly longer than the roughing, 2 and a half hours compared to 20 minutes. Below are the tool paths and resulting IPWs during the machining process.







Projected Output & Production Cost per Part

Output Predictions

With a total machining time of 2 hours and 45 minutes per part and additional time for fixture setup, pre-processing, post-processing, and tool changes, production is estimated at two parts per workday. Assuming a 9-hour workday, this results in approximately 520 parts produced per year.

Production Cost per Part

The total production cost per part is calculated to be $35.24, which includes a raw material cost of $31.68 per part, a machine cost of $3.06 per part based on a $31,800 machine amortized over a 20-year life, and a fixture cost of $0.50 per part based on a $1,300 fixture amortized over five years. Labor costs are not included, as the labor is assumed to be performed by the owner.

Sales & Revenue

The gas cap is priced at $70 per part, which is comparable to similar products on the market and reflects its high-quality material and finish. With an estimated production volume of 520 parts per year, total annual sales are projected to be $36,400. After subtracting the production cost of $35.24 per part, the estimated yearly profit is approximately $18,075.20, indicating that the product can generate a solid return at the planned production rate. Below is the math which has been used to calculate revenue.

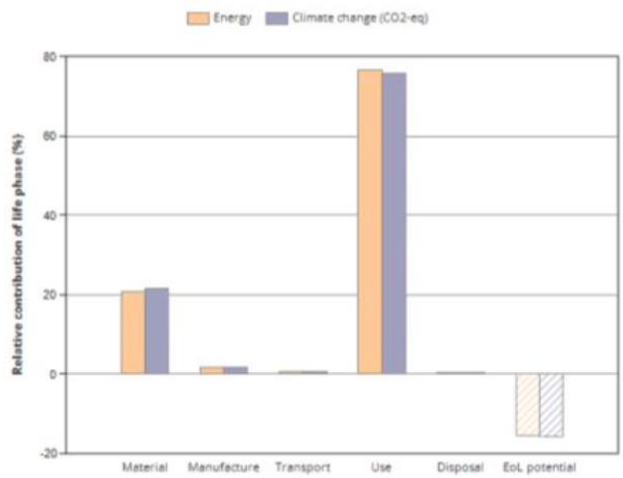
$$\left(-35.24(520)+70(520)\right)$$

$$= \boxed{18075.2}$$

Sustainability Impacts

Using GRANTA, my analysis shows that most of the environmental impact occurs during the use phase, since the gas cap is installed on an automobile that operates over many years. Material production and manufacturing contribute a smaller portion of the total impact due to the part’s relatively low mass and simple geometry, while transportation and packaging have minimal effects because the component is lightweight. End-of-life disposal contributes very little, and recycling provides a small environmental benefit that helps offset earlier impacts. Overall, the gas cap has a relatively low environmental footprint compared to larger automotive components, with its impact largely tied to vehicle operation rather than the part itself. The summary of my GRANTA analysis is posted here:

SUMMARY:



Energy details

Climate change (CO2-eq) details

Phase	Energy (kcal)	Energy (%)	Climate change (CO2-eq) (lb)	Climate change (CO2-eq) (%)
Material	6.3e+04	20.7	41.6	21.5
Manufacture	5.06e+03	1.7	3.5	1.8
Transport	2.4e+03	0.8	1.46	0.8
Use	2.33e+05	76.7	146	75.8
Disposal	356	0.1	0.23	0.1
Total (for first life)	**3.04e+05**	**100**	**193**	**100**
End of life potential	-4.73e+04		-30.4	

Project Summary & Outlook

I think that this product is viable to create via a CNC mill and may have potential to build a business around given the variables explained. In conclusion, the product has value to a particular market which is large enough to support a small business. The CNC mill chosen will work well for machining this part and can create accurate IPWs. Machining this product may generate a net profit of around $18,000.00 within a year. This can be done during normal operating hours of 9am to 5pm, producing 2 parts per day. The biggest setback of this plan as it stands right now, is the machining time, and the material cost. If the machine time may be perfected to increase the number of parts per day, this may justify even hiring labor. Also, if we were able to find inexpensive material, this could offset production cost as well.